



UNIVERSITI MALAYSIA TERENGGANU

FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH HONORS

SEMESTER II 2025/2026

CSA3023

WEB-BASED APPLICATION DEVELOPMENT

LAB 6:

SESSION MANAGEMENT, FILE UPLOAD & MVC

Prepared by:

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Prepared for:

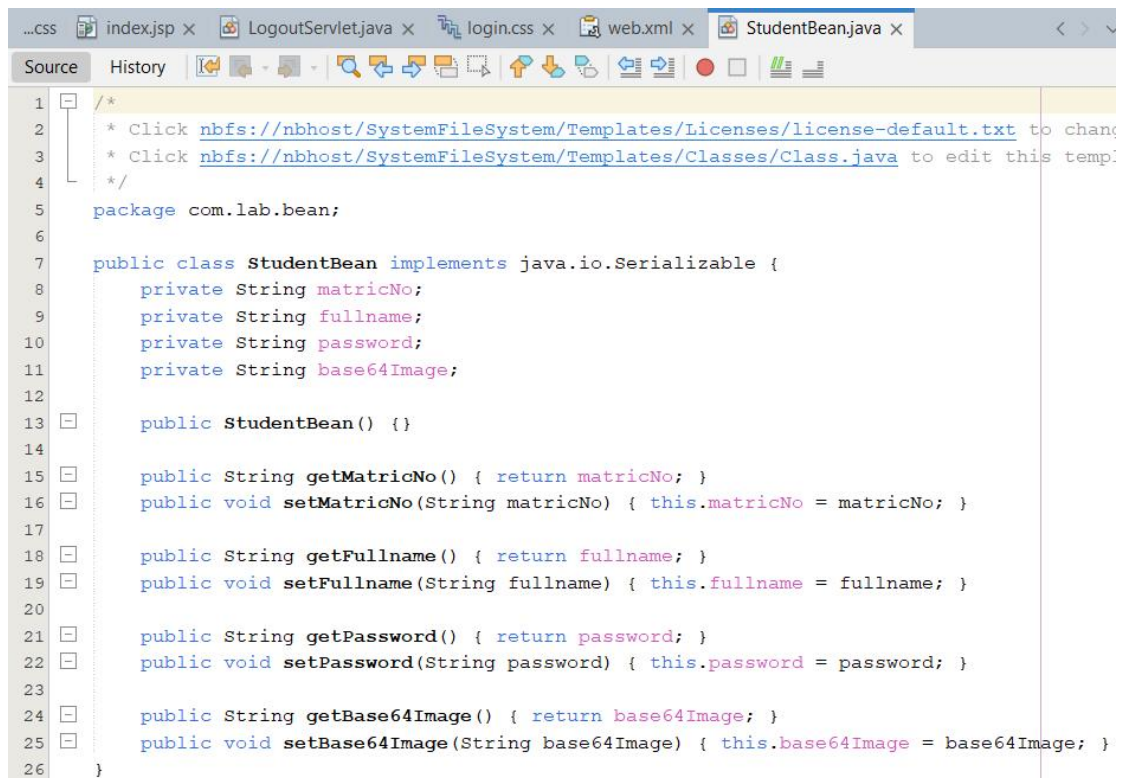
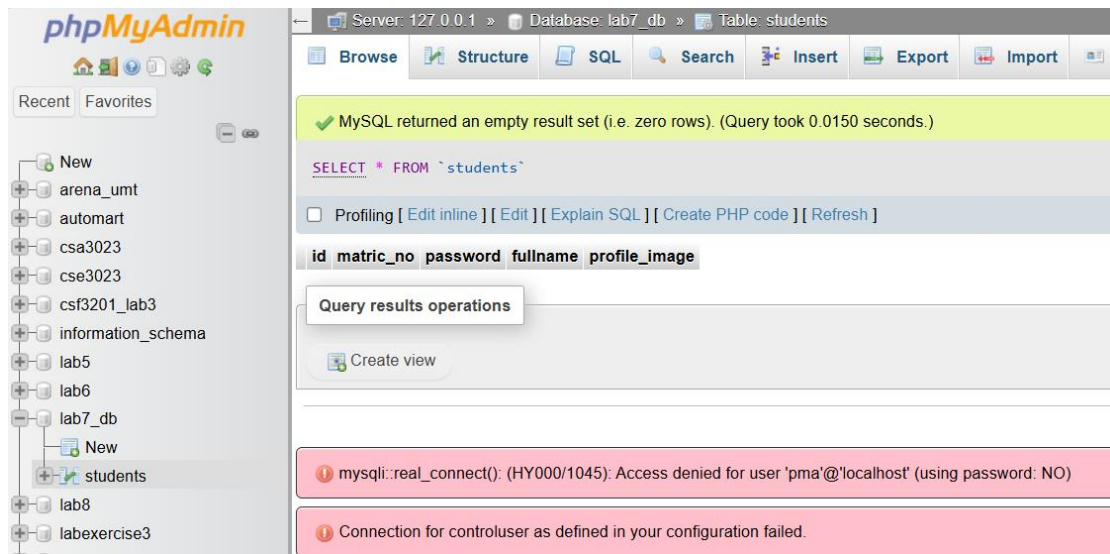
SIR. MOHAMAD ARIZAL SHAMSIL MAT RIFFIN

Date:

19 MAY 2026

Task 1: Building the Model (JavaBean & DAO)

Objective: To encapsulate data using a JavaBean and handle all database SQL logic inside a separate Data Access Object (DAO) class



```
...va dashboard.jsp x UpdateProfileServlet.java x dashboard.css x index.jsp x LogoutServlet.java x login.css x web.xml x S
Source History
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
package com.lab.dao;

import com.lab.bean.StudentBean;
import java.io.ByteArrayOutputStream;
import java.io.InputStream;
import java.sql.Blob;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Base64;

public class StudentDAO {

    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection("jdbc:mysql://localhost:3306/lab7_db",
            "root", "");
    }

    // --- REGISTER LOGIC ---
    public boolean registerStudent(StudentBean student, InputStream imageStream) {
        try (Connection conn = getConnection()) {
            String sql =
                "INSERT INTO students (matric_no, password, fullname, profile_image) VALUES(?, ?, ?, ?)";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setString(1, student.getMatricNo());
            pstmt.setString(2, student.getPassword());
            pstmt.setString(3, student.getFullname());

            pstmt.setBlob(4, imageStream); // Set BLOB data

            return pstmt.executeUpdate() > 0;
        } catch (Exception e) {
            e.printStackTrace();
            return false;
        }
    }

    // --- LOGIN LOGIC ---
    public StudentBean loginStudent(String matricNo, String password) {
        StudentBean student = null;
        try (Connection conn = getConnection()) {
            String sql = "SELECT * FROM students WHERE matric_no = ? AND password = ?";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setString(1, matricNo);

            pstmt.setString(2, password);
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                student = new StudentBean();
                student.setMatricNo(rs.getString("matric_no"));
                student.setFullname(rs.getString("fullname"));

                // Fetch BLOB and convert to Base64 String
                Blob blob = rs.getBlob("profile_image");
                if (blob != null) {
                    InputStream inputStream = blob.getBinaryStream();
                    ByteArrayOutputStream outputStream = new ByteArrayOutputStream();
                    byte[] buffer = new byte[4096];
                    int bytesRead = -1;
                    while ((bytesRead = inputStream.read(buffer)) != -1) {
                        outputStream.write(buffer, 0, bytesRead);
                    }
                }
            }
        }
    }
}
```

```

66         }
67         byte[] imageBytes = outputStream.toByteArray();
68         student.setBase64Image(Base64.getEncoder().encodeToString(imageBytes));
69     }
70 }
71 } catch (Exception e) {
72     e.printStackTrace();
73 }
74     return student;
75 }
76 // --- DELETE LOGIC ---
77 public boolean deleteStudent(String matricNo) {
78     try (Connection conn = getConnection()) {
79         String sql = "DELETE FROM students WHERE matric_no = ?";
80         PreparedStatement pstmt = conn.prepareStatement(sql);
81         pstmt.setString(1, matricNo);
82         return pstmt.executeUpdate() > 0;
83     } catch (Exception e) {
84         e.printStackTrace();
85         return false;
86     }
87 }
88 }

```

Task 2: The Unified Controller (UserServlet)

Objective: To consolidate multiple operations (Login, Logout, Register, Delete Account) into a single Servlet that directs traffic to the DAO based on an action parameter.

```

...jsp UpdateProfileServlet.java x dashboard.css x index.jsp x LogoutServlet.java x login.css x web.xml x
Source History
1 package com.lab.controller;
2
3 import com.lab.bean.StudentBean;
4 import com.lab.dao.StudentDAO;
5
6 import java.io.IOException;
7 import java.io.InputStream;
8
9 import javax.servlet.ServletException;
10 import javax.servlet.annotation.MultipartConfig;
11 import javax.servlet.http.HttpServlet;
12 import javax.servlet.http.HttpServletRequest;
13 import javax.servlet.http.HttpServletResponse;
14 import javax.servlet.http.HttpSession;
15 import javax.servlet.http.Part;
16
17 @MultipartConfig(maxFileSize = 16177215)
18
19 public class UserServlet extends HttpServlet {
20
21     private StudentDAO studentDAO = new StudentDAO();
22
23     @Override
24     protected void doPost(HttpServletRequest request,
25         HttpServletResponse response)
26         throws ServletException, IOException {
27
28         response.setContentType("text/html;charset=UTF-8");
29
30         String action = request.getParameter("action");
31

```

```

32 // REGISTER
33 if ("register".equals(action)) {
34
35     StudentBean newStudent = new StudentBean();
36
37     newStudent.setMatricNo(
38         request.getParameter("matricNo"));
39
40     newStudent.setFullname(
41         request.getParameter("fullname"));
42
43     newStudent.setPassword(
44         request.getParameter("password"));
45
46     InputStream inputStream = null;
47
48     Part filePart = request.getPart("profileImage");
49
50     if (filePart != null && filePart.getSize() > 0) {
51
52         inputStream = filePart.getInputStream();
53     }
54
55     boolean isSuccess =
56         studentDAO.registerStudent(newStudent, inputStream);
57
58     if (isSuccess) {
59
60         response.getWriter().println(
61             "Registration Successful! "
62             + "<a href='login.html'>Login here</a>"
63         );
64
65     } else {
66
67         response.getWriter().println(
68             "Registration Failed!"
69         );
70     }
71
72 }
73
74 // LOGIN
75 else if ("login".equals(action)) {
76
77     String matric =
78         request.getParameter("matricNo");
79
80     String pass =
81         request.getParameter("password");
82
83     StudentBean student =
84         studentDAO.loginStudent(matric, pass);
85
86     if (student != null) {
87
88         HttpSession session =
89             request.getSession();
90
91         session.setAttribute(
92             "loggedUser",
93             student
94         );
95
96         response.sendRedirect("dashboard.jsp");

```



```

98         } else {
99
100             response.getWriter().println(
101                 "Invalid Credentials! "
102                 + "<a href='login.html'>Try Again</a>"
103             );
104         }
105     }
106 }
107
108 @Override
109 protected void doGet(HttpServletRequest request,
110     HttpServletResponse response)
111     throws ServletException, IOException {
112
113     String action =
114         request.getParameter("action");
115
116     HttpSession session =
117         request.getSession(false);
118
119     // LOGOUT
120     if ("logout".equals(action)) {
121
122         if (session != null) {
123
124             session.invalidate();
125         }
126
127         response.sendRedirect("login.html");
128     }
129
130     // DELETE ACCOUNT
131     else if ("delete".equals(action)) {
132
133         if (session != null
134             && session.getAttribute("loggedUser") != null) {
135
136             StudentBean user =
137                 (StudentBean) session.getAttribute("loggedUser");
138
139             studentDAO.deleteStudent(
140                 user.getMatricNo());
141
142             session.invalidate();
143         }
144
145         response.sendRedirect("register.html");
146     }
147 }
148 }

```

Task 3: The Views and Session Security

Objective: To create the HTML forms containing hidden action fields and a secure dashboard using JSP

```
...css web.xml x StudentBean.java x StudentDAO.java x UserServlet.java x register.html x
Source History
1 <!DOCTYPE html>
2 <!--
3 Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this
4 Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this template
5 -->
6 <html>
7 <head><title>Student Registration</title></head>
8 <body>
9 <h2>Register New Student</h2>
10 <form action="UserServlet" method="POST" enctype="multipart/form-data">
11 <input type="hidden" name="action" value="register">
12
13 Matric No: <input type="text" name="matricNo" required><br><br>
14 Full Name: <input type="text" name="fullname" required><br><br>
15 Password: <input type="password" name="password" required><br><br>
16 Profile Picture: <input type="file" name="profileImage" accept="image/*" required>
17 <br>
18 <br>
19 <input type="submit" value="Register">
20 </form>
21 </body>
22 </html>
```

```
...ml StudentBean.java x StudentDAO.java x UserServlet.java x register.html x login.html x
Source History
1 <!DOCTYPE html>
2 <!--
3 Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to ch
4 Click nbfs://nbhost/SystemFileSystem/Templates/JSP\_Servlet/Html.html to edit this
5 -->
6 <html>
7 <body>
8 <h2>Student Login</h2>
9 <form action="UserServlet" method="POST">
10 <input type="hidden" name="action" value="login">
11 Matric No: <input type="text" name="matricNo" required><br><br>
12 Password: <input type="password" name="password" required><br><br>
13
14 <input type="submit" value="Login">
15 </form>
16 </body>
17 </html>
18
```

```
...css index.jsp X LogoutServlet.java X login.css X web.xml X StudentBean.java X StudentDAO.java X UserServlet.java
Source History
1 <!--
2 Document : dashboard
3 Created on : 19 May 2026, 4:33:12 pm
4 Author : syaheera
5 -->
6
7 <%@page contentType="text/html" pageEncoding="UTF-8"%>
8 <%@page import="com.lab.bean.StudentBean"%>
9 <!DOCTYPE html>
10 <html lang="en">
11 <head>
12 <meta charset="UTF-8">
13 <meta name="viewport" content="width=device-width, initial-scale=1">
14 <title>Student Dashboard</title>
15 <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
16 rel="stylesheet">
17
18 <style>
19 /* Custom CSS minor untuk saiz dan bingkai gambar sahaja */
20 .profile-img {
21 width: 160px;
22 height: 160px;
23 object-fit: cover;
24 border: 5px solid #198754; /* Warna hijau success Bootstrap */
25 }
26 </style>
27 </head>
28 <body class="bg-light">
29 <%
30 // 1. Session Verification (Security Check)
31 response.setContentType("text/html;charset=UTF-8");
32 StudentBean user = (StudentBean) session.getAttribute("loggedUser");
33 if(user == null) {
34 response.sendRedirect("login.html");
35 return;
36 }
37 <%>
38 <div class="container d-flex justify-content-center align-items-center"
39 style="min-height: 100vh;">
40 <div class="card shadow-lg border-0 rounded-4 p-4 p-md-5 text-center"
41 style="max-width: 600px; width: 100%;">
42 <div class="card-body">
43 <h2 class="fw-bold text-dark mb-2">Welcome, <%= user.getFullname() %>!</h2>
45 <p class="text-muted fs-5 mb-4">Matric Number: <strong class="text-secondary"><div class="d-flex justify-content-center gap-3 flex-wrap">
46 <a href="UserServlet?action=logout" class="btn btn-dark px-4 py-2
47 fw-semibold">Logout</a>
48 <a href="UserServlet?action=delete"
49 onclick="return confirm('WARNING: Are you sure you want to delete
50 your account permanently?');">
51 class="btn btn-danger px-4 py-2 fw-semibold">Delete Account</a>
52 </div>
53 </div>
54 </div>
55 </div>
56 <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/
57 bootstrap.bundle.min.js"></script>
58 </body>
59 </html>
```

Task 4: Debugging and Viewing Server Logs

Objective: To learn how to trace and troubleshoot backend errors (e.g., HTTP 500) by inspecting the server logs and Output window in NetBeans.

Exercise: Student Subject Management System (CRUD)

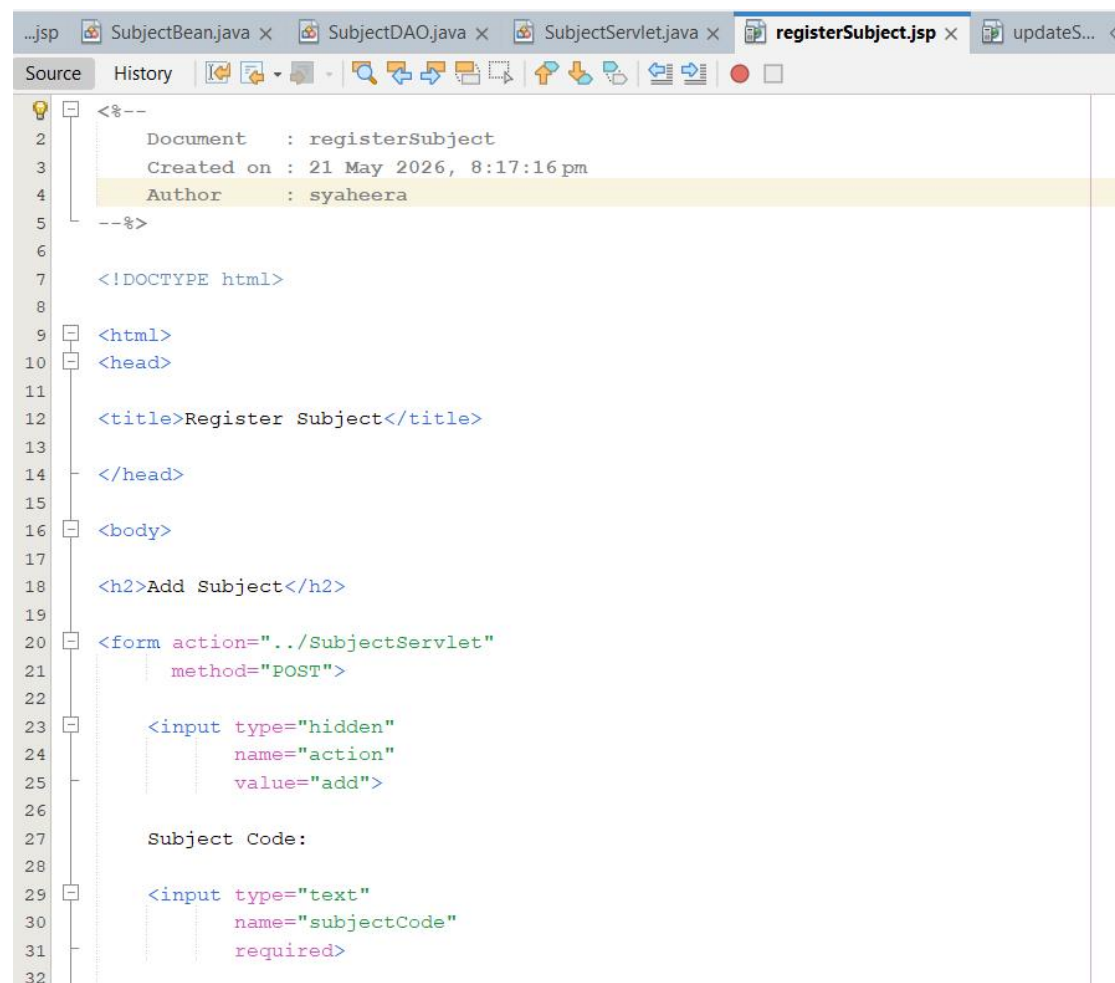
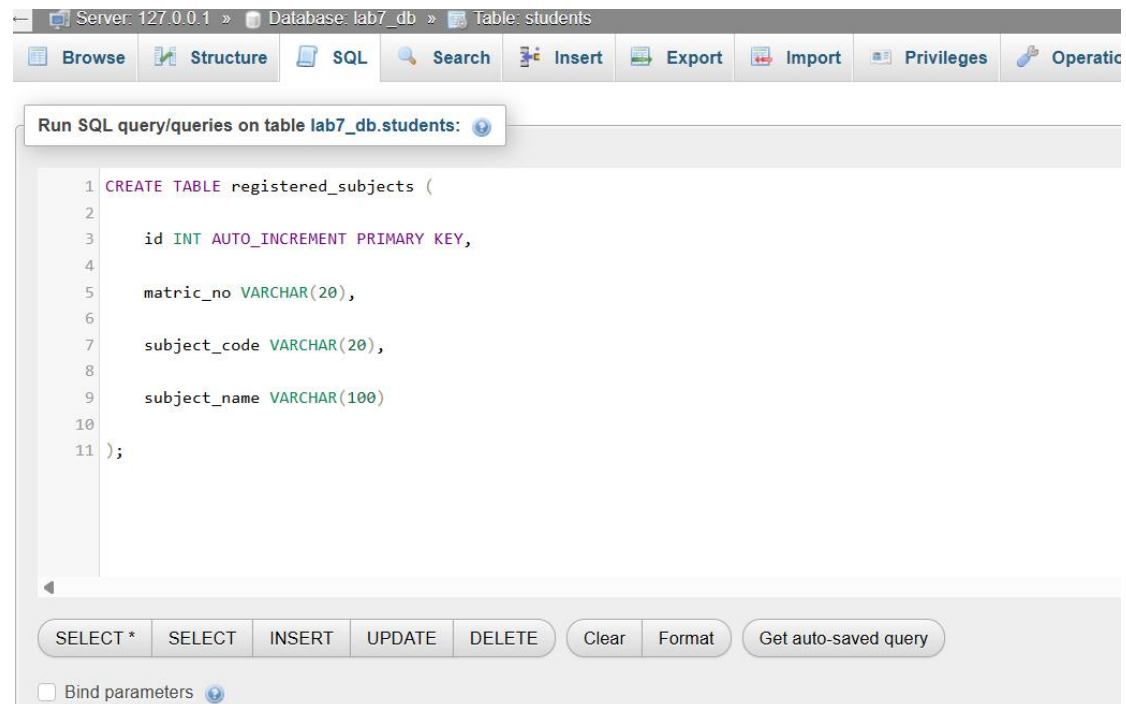
Based on the MVC architecture and Session Management techniques learned in this lab, you are required to extend the current application by developing a Subject Management System. This system must allow a logged-in student to manage their academic subjects through a full CRUD cycle.

System Requirements:

1. Database Design:

- Create a table named `registered_subjects`.

- It must at least contain: id (PK), matric_no (to link with the logged-in user), and subject_name or subject_code.



```

33     <br><br>
34
35     Subject Name:
36
37     <input type="text"
38           name="subjectName"
39           required>
40
41     <br><br>
42
43     <input type="submit"
44           value="Add Subject">
45
46 </form>
47
48 <br>
49
50 <a href="../SubjectServlet?action=view">
51
52 View Subjects
53
54 </a>
55
56 </body>
57 </html>

```

...va SubjectServlet.java x registerSubject.jsp x updateSubject.jsp x viewSubjects.jsp x

Source History

```

1 <!--
2     Document : updateSubject
3     Created on : 21 May 2026, 8:17:58 pm
4     Author : User
5 -->
6
7 <%@page import="com.lab.bean.SubjectBean"%>
8
9 <%
10
11     SubjectBean s =
12     (SubjectBean)
13     request.getAttribute("subject");
14
15 %>
16
17 <!DOCTYPE html>
18
19 <html>
20 <head>
21
22 <title>Update Subject</title>
23
24 </head>
25
26 <body>
27
28 <h2>Update Subject</h2>
29
30 <form action="../SubjectServlet"
31       method="POST">
32

```

```
33 <input type="hidden"
34       name="action"
35       value="update">
36
37 <input type="hidden"
38       name="id"
39       value="<%= s.getId() %>">
40
41 Subject Code:
42
43 <input type="text"
44       name="subjectCode"
45       value="<%= s.getSubjectCode() %>">
46
47 <br><br>
48
49 Subject Name:
50
51 <input type="text"
52       name="subjectName"
53       value="<%= s.getSubjectName() %>">
54
55 <br><br>
56
57 <input type="submit"
58       value="Update">
59
60 </form>
61
62 </body>
63 </html>
64
```

```
...va SubjectServlet.java x registerSubject.jsp x updateSubject.jsp x viewSubjects.jsp x
Source History
1 <!--
2     Document    : viewSubjects
3     Created on  : 21 May 2026, 8:17:39 pm
4     Author     : syaheera
5 -->
6
7 <%@page import="java.util.List"%>
8 <%@page import="com.lab.bean.SubjectBean"%>
9
10 <!DOCTYPE html>
11
12 <html>
13 <head>
14
15 <title>View Subjects</title>
16
17 </head>
18
19 <body>
20
21 <h2>Subject List</h2>
22
23 <a href="registerSubject.jsp">
24
25     Add New Subject
26
27 </a>
28
29 <br><br>
30
31 <table border="1"
32     cellpadding="10">
```

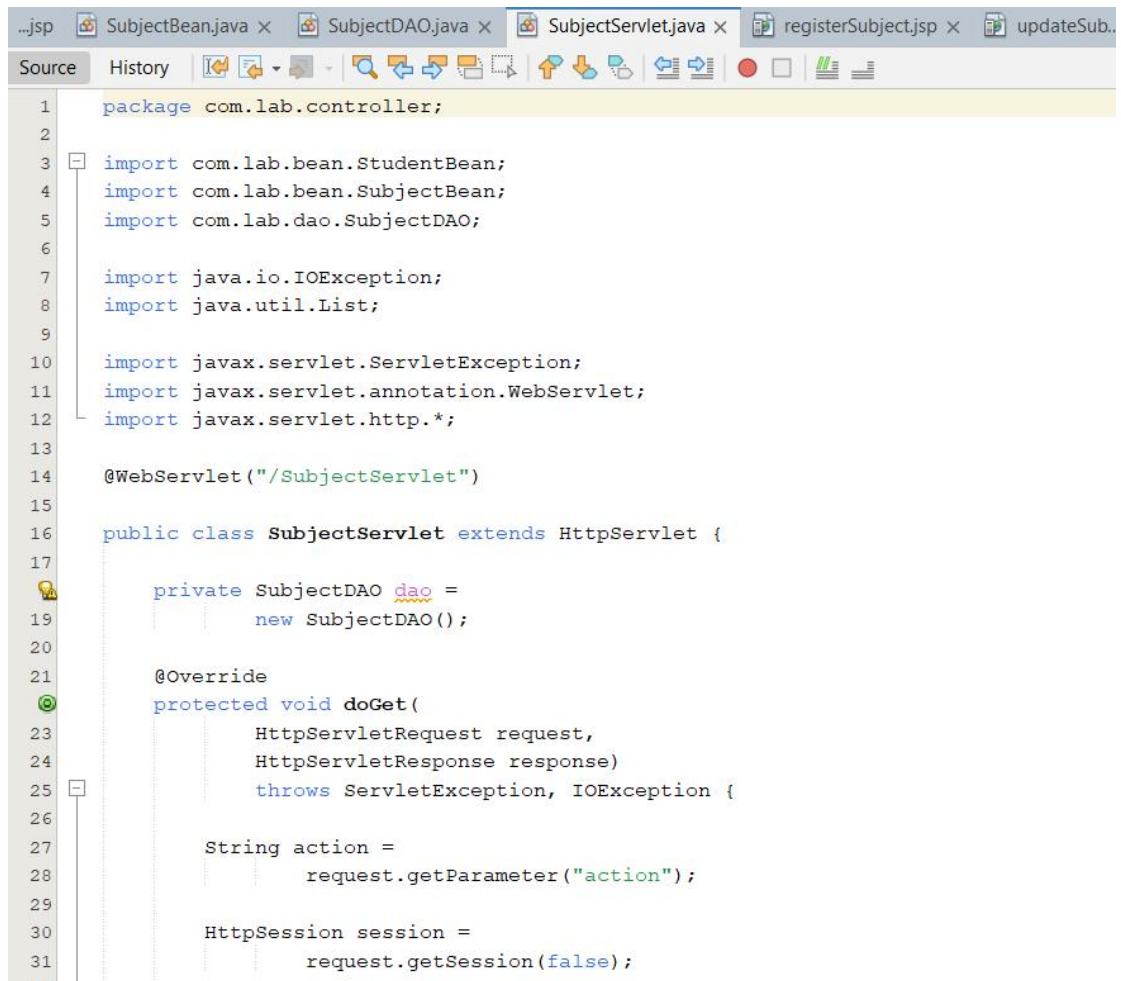
```

33
34 <tr>
35
36 <th>ID</th>
37 <th>Subject Code</th>
38 <th>Subject Name</th>
39 <th>Action</th>
40
41 </tr>
42
43 <%
44
45 List<SubjectBean> list =
46 (List<SubjectBean>)
47 request.getAttribute("subjectList");
48
49 for (SubjectBean s : list){
50
51 %>
52
53 <tr>
54
55 <td><%= s.getId() %></td>
56
57 <td><%= s.getSubjectCode() %></td>
58
59 <td><%= s.getSubjectName() %></td>
60
61 <td>
62
63 <a href="../SubjectServlet?action=edit&id=<%= s.getId() %>">
64
65 Edit
66
67 </a>
68
69 |
70
71 <a href="../SubjectServlet?action=delete&id=<%= s.getId() %>"
72 onclick="return confirm('Delete this subject?')">
73
74 Delete
75
76 </a>
77
78 </td>
79
80 </tr>
81
82 <%
83 }
84 %>
85
86 </table>
87
88 </body>
89 </html>

```



```
...jsp SubjectBean.java x SubjectDAO.java x SubjectServlet.java x registerSubject.jsp x updateS
Source History
1 package com.lab.bean;
2
3 public class SubjectBean {
4
5     private int id;
6     private String matricNo;
7     private String subjectCode;
8     private String subjectName;
9
10    public SubjectBean() {
11    }
12
13    public int getId() {
14        return id;
15    }
16
17    public void setId(int id) {
18        this.id = id;
19    }
20
21    public String getMatricNo() {
22        return matricNo;
23    }
24
25    public void setMatricNo(String matricNo) {
26        this.matricNo = matricNo;
27    }
28
29    public String getSubjectCode() {
30        return subjectCode;
31    }
32
33    public void setSubjectCode(String subjectCode) {
34        this.subjectCode = subjectCode;
35    }
36
37    public String getSubjectName() {
38        return subjectName;
39    }
40
41    public void setSubjectName(String subjectName) {
42        this.subjectName = subjectName;
43    }
44 }
```



```
1 package com.lab.controller;
2
3 import com.lab.bean.StudentBean;
4 import com.lab.bean.SubjectBean;
5 import com.lab.dao.SubjectDAO;
6
7 import java.io.IOException;
8 import java.util.List;
9
10 import javax.servlet.ServletException;
11 import javax.servlet.annotation.WebServlet;
12 import javax.servlet.http.*;
13
14 @WebServlet("/SubjectServlet")
15
16 public class SubjectServlet extends HttpServlet {
17
18     private SubjectDAO dao =
19         new SubjectDAO();
20
21     @Override
22     protected void doGet(
23         HttpServletRequest request,
24         HttpServletResponse response)
25         throws ServletException, IOException {
26
27         String action =
28             request.getParameter("action");
29
30         HttpSession session =
31             request.getSession(false);
```

```

32
33     StudentBean user =
34         (StudentBean)
35         session.getAttribute("loggedUser");
36
37     String matricNo =
38         user.getMatricNo();
39
40     // VIEW
41     if(action.equals("view")){
42
43         List<SubjectBean> list =
44             dao.getAllSubjects(matricNo);
45
46         request.setAttribute(
47             "subjectList",
48             list
49         );
50
51         request.getRequestDispatcher(
52             "subject/viewSubjects.jsp"
53         ).forward(request, response);
54     }
55
56     // DELETE
57     else if(action.equals("delete")){
58
59         int id =
60             Integer.parseInt(
61                 request.getParameter("id")
62             );
63
64         dao.deleteSubject(id);
65
66         response.sendRedirect(
67             "SubjectServlet?action=view"
68         );
69     }
70
71     // EDIT
72     else if(action.equals("edit")){
73
74         int id =
75             Integer.parseInt(
76                 request.getParameter("id")
77             );
78
79         SubjectBean subject =
80             dao.getSubjectById(id);
81
82         request.setAttribute(
83             "subject",
84             subject
85         );
86
87         request.getRequestDispatcher(
88             "subject/updateSubject.jsp"
89         ).forward(request, response);
90     }
91 }
92
93 @Override

```

```

95     protected void doPost(
96         HttpServletRequest request,
97         HttpServletResponse response)
98         throws ServletException, IOException {
99
100         String action =
101             request.getParameter("action");
102
103         HttpSession session =
104             request.getSession(false);
105
106         StudentBean user =
107             (StudentBean)
108             session.getAttribute("loggedUser");
109
110         String matricNo =
111             user.getMatricNo();
112
113         // ADD
114         if(action.equals("add")){
115
116             SubjectBean s =
117                 new SubjectBean();
118
119             s.setMatricNo(matricNo);
120
121             s.setSubjectCode(
122                 request.getParameter("subjectCode")
123             );
124
125             s.setSubjectName(
126                 request.getParameter("subjectName")
127             );
128
129             dao.addSubject(s);
130
131             response.sendRedirect(
132                 "SubjectServlet?action=view"
133             );
134         }
135
136         // UPDATE
137         else if(action.equals("update")){
138
139             SubjectBean s =
140                 new SubjectBean();
141
142             s.setId(
143                 Integer.parseInt(
144                     request.getParameter("id")
145                 )
146             );
147
148             s.setSubjectCode(
149                 request.getParameter("subjectCode")
150             );
151
152             s.setSubjectName(
153                 request.getParameter("subjectName")
154             );
155
156             dao.updateSubject(s);

```

```

156
157         response.sendRedirect(
158             "SubjectServlet?action=view"
159         );
160     }
161 }
162

```

...ml login.html x dashboard.jsp x SubjectBean.java x SubjectDAO.java x SubjectServlet.java...

Source History

```

1 package com.lab.dao;
2
3 import com.lab.bean.SubjectBean;
4
5 import java.sql.Connection;
6 import java.sql.DriverManager;
7 import java.sql.PreparedStatement;
8 import java.sql.ResultSet;
9
10 import java.util.ArrayList;
11 import java.util.List;
12
13 public class SubjectDAO {
14
15     private Connection getConnection() throws Exception {
16
17         Class.forName("com.mysql.cj.jdbc.Driver");
18
19         return DriverManager.getConnection(
20             "jdbc:mysql://localhost:3306/lab7_db",
21             "root",
22             "admin"
23         );
24     }
25
26     // ADD SUBJECT
27     public boolean addSubject(SubjectBean subject) {
28
29         try (Connection conn = getConnection()) {
30
31             String sql =

```

```

32         "INSERT INTO registered_subjects "
33         + "(matric_no, subject_code, subject_name) "
34         + "VALUES (?, ?, ?)";
35
36         PreparedStatement ps =
37             conn.prepareStatement(sql);
38
39         ps.setString(1, subject.getMatricNo());
40         ps.setString(2, subject.getSubjectCode());
41         ps.setString(3, subject.getSubjectName());
42
43         return ps.executeUpdate() > 0;
44
45     } catch (Exception e) {
46
47         e.printStackTrace();
48         return false;
49     }
50 }
51
52 // VIEW SUBJECT
53 public List<SubjectBean> getAllSubjects(
54     String matricNo) {
55
56     List<SubjectBean> list =
57         new ArrayList<>();
58
59     try (Connection conn = getConnection()) {
60
61         String sql =
62             "SELECT * FROM registered_subjects "
63             + "WHERE matric_no=?";
64
65         PreparedStatement ps =
66             conn.prepareStatement(sql);
67
68         ps.setString(1, matricNo);
69
70         ResultSet rs =
71             ps.executeQuery();
72
73         while(rs.next()){
74
75             SubjectBean s =
76                 new SubjectBean();
77
78             s.setId(
79                 rs.getInt("id")
80             );
81
82             s.setMatricNo(
83                 rs.getString("matric_no")
84             );
85
86             s.setSubjectCode(
87                 rs.getString("subject_code")
88             );
89
90             s.setSubjectName(
91                 rs.getString("subject_name")
92             );
93

```



```

94         list.add(s);
95     }
96
97     } catch (Exception e) {
98
99         e.printStackTrace();
100     }
101
102     return list;
103 }
104
105 // DELETE SUBJECT
106 public boolean deleteSubject(int id){
107
108     try(Connection conn = getConnection()){
109
110         String sql =
111             "DELETE FROM registered_subjects "
112             + "WHERE id=?";
113
114         PreparedStatement ps =
115             conn.prepareStatement(sql);
116
117         ps.setInt(1, id);
118
119         return ps.executeUpdate() > 0;
120
121     } catch (Exception e){
122
123         e.printStackTrace();
124         return false;

```

```

125     }
126 }
127
128 // GET SUBJECT BY ID
129 public SubjectBean getSubjectById(int id){
130
131     SubjectBean s = null;
132
133     try(Connection conn = getConnection()){
134
135         String sql =
136             "SELECT * FROM registered_subjects "
137             + "WHERE id=?";
138
139         PreparedStatement ps =
140             conn.prepareStatement(sql);
141
142         ps.setInt(1, id);
143
144         ResultSet rs =
145             ps.executeQuery();
146
147         if(rs.next()){
148
149             s = new SubjectBean();
150
151             s.setId(
152                 rs.getInt("id")
153             );
154
155             s.setSubjectCode(

```

```

156         rs.getString("subject_code")
157     );
158
159     s.setSubjectName(
160         rs.getString("subject_name")
161     );
162 }
163
164 } catch (Exception e){
165     e.printStackTrace();
166 }
167
168
169 return s;
170 }
171
172 // UPDATE SUBJECT
173 public boolean updateSubject(
174     SubjectBean subject){
175
176     try(Connection conn = getConnection()){
177
178         String sql =
179             "UPDATE registered_subjects "
180             + "SET subject_code=?, "
181             + "subject_name=? "
182             + "WHERE id=?";
183
184         PreparedStatement ps =
185             conn.prepareStatement(sql);
186
187         ps.setString(
188             1,
189             subject.getSubjectCode()
190         );
191
192         ps.setString(
193             2,
194             subject.getSubjectName()
195         );
196
197         ps.setInt(
198             3,
199             subject.getId()
200         );
201
202         return ps.executeUpdate() > 0;
203
204     } catch (Exception e){
205         e.printStackTrace();
206         return false;
207     }
208 }
209
210 }

```

localhost:8081/Excercise7/register.html

Register

Matric No:

Full Name:

Password:

[Login Here](#)

localhost:8081/Excercise7/UserServlet

Register Failed

localhost:8081/Excercise7/login.html

Login

Matric No:

Password:

[Register Here](#)

localhost:8081/Excercise7/UserServlet

Invalid Login